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Prime CAT 04 2022 DILR

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Section-1

Sec 1

Directions for questions 1 to 6: Answer the questions on the basis of the information given below.

As listed in the table below, 8 teams participated in the GK Quiz on International Yoga Day consisting of 8 rounds. There were 5 participating teams in each round. Teams A to E participated in Round 1, Teams B to F in Round 2, ... , Teams D to H in Round 4, Teams E to H and A in Round 5, Teams F to H, A and B in Round 6 and so on.

The top three performing teams in each round were awarded 5, 3 and 2 points respectively. There was no tie in any of the 8 rounds. The table below shows the total points scored by 8 teams after Round 5 and Round 8.

Team	Scores after round 5	Scores after round 8
A	7	15
B	4	7
C	2	6
D	6	11
E	3	3
F	8	8
G	13	16
H	7	14

The following information is also known.
(i) Team F did not score consecutively in any two rounds from Rounds 1 to 5.
(ii) Both teams D and F scored in a round from Rounds 1 to 5.
(iii) No team scored in all the three rounds from Rounds 6 to 8.
(iv) Only two teams scored in two consecutive rounds from Rounds 6 to 8.

Q.1 [11831809]
What was the sum of the scores of teams D, E and F after Round 4?

Solution:
Correct Answer : 17

Answer key/Solution

Let's say R1, R2, …, R8 represent the rounds 1 to 8.

Step 1:

From the given information, first we calculate the points scored by each team from rounds 6 to 8. Points scored by team A from rounds 6 to 8 = 15 – 7 = 8. Similarly, we can find for the other teams. Then, we find which teams participated in the given rounds. Team A participated in rounds R1, R5, R6, R7, R8. Similarly, we can find for other teams.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5		8	R6, R7, R8		15
B	4	R1, R2		3	R6, R7, R8		7
C	2	R1, R2, R3		4	R7, R8		6
D	6	R1, R2, R3, R4		5	R8		11
E	3	R1, R2, R3, R4, R5		0	----------		3
F	8	R2, R3, R4, R5		0	R6		8
G	13	R3, R4, R5		3	R6, R7		16
H	7	R4, R5		7	R6, R7, R8		14

Step 2:

Now let us calculate the points scored by the teams in different rounds.

From condition (i), team F scored in two rounds out of four rounds R2, R3, R4, R5.

From condition (iii), all teams scored in a maximum of two rounds from rounds 6 to 8.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5	2, 5	8	R6, R7, R8	3, 5	15
B	4	R1, R2	2, 2	3	R6, R7, R8	3	7
C	2	R1, R2, R3	2	4	R7, R8	2, 2	6
D	6	R1, R2, R3, R4	3, 3	5	R8	5	11
E	3	R1, R2, R3, R4, R5	3	0	----------	0	3
F	8	R2, R3, R4, R5	3, 5	0	R6	0	8
G	13	R3, R4, R5	3, 5, 5	3	R6, R7	3	16
H	7	R4, R5	2, 5	7	R6, R7, R8	2, 5	14

Step 3:

From condition (i), team F scored 5 and 2 points in rounds R2 and R4 respectively.

From condition (ii), both teams D and F scored in round R2.

From condition (iii), all teams scored in a maximum of two rounds out of R6, R7 and R8 rounds.

From condition (iv), only team C and H scored in two consecutive rounds from rounds 6 to 8.

From the above table, team B scored 2, 2 points in rounds R1 and R2. So team A scored 5, 2 points in rounds R1 and R5 respectively. Therefore, C scored 2 points in round R3.

So, team H scored 2, 5 points in rounds R4 and R5 respectively. Hence, team G scored 3, 5, 5 points in rounds R5, R4 and R3 respectively. Similarly, we can fill for the other teams.

Hence, the final table can be shown as:

Round No.	5 Points	3 Points	2 Points
R1	A	D/E	B
R2	F	D	B
R3	G	E/D	C
R4	G	F	H
R5	H	G	A
R6	A	B/G	H
R7	H	G/B	C

R8	D	A	C

The sum of the scores of teams D, E and F after Round 4 = (3 + 3) + 3 + (5 + 3) = 17.

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The following information is also known.
(i) Team F did not score consecutively in any two rounds from Rounds 1 to 5.
(ii) Both teams D and F scored in a round from Rounds 1 to 5.
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(iv) Only two teams scored in two consecutive rounds from Rounds 6 to 8.

Q.2 [11831809]
Which of the following teams were in the last four positions after Round 3?

1 ○ B, C, D, H

2 ○ C, D, E, G

3 ○ B, C, E, H

4 ○ A, C, E, F

Solution:
Correct Answer : 3

Answer key/Solution

Let's say R1, R2, …, R8 represent the rounds 1 to 8.

Step 1:

From the given information, first we calculate the points scored by each team from rounds 6 to 8. Points scored by team A from rounds 6 to 8 = 15 – 7 = 8. Similarly, we can find for the other teams. Then, we find which teams participated in the given rounds. Team A participated in rounds R1, R5, R6, R7, R8. Similarly, we can find for other teams.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5		8	R6, R7, R8		15
B	4	R1, R2		3	R6, R7, R8		7
C	2	R1, R2, R3		4	R7, R8		6
D	6	R1, R2, R3, R4		5	R8		11
E	3	R1, R2, R3, R4, R5		0	----------		3
F	8	R2, R3, R4, R5		0	R6		8
G	13	R3, R4, R5		3	R6, R7		16
H	7	R4, R5		7	R6, R7, R8		14

Step 2:

Now let us calculate the points scored by the teams in different rounds.

From condition (i), team F scored in two rounds out of four rounds R2, R3, R4, R5.

From condition (iii), all teams scored in a maximum of two rounds from rounds 6 to 8.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5	2, 5	8	R6, R7, R8	3, 5	15
B	4	R1, R2	2, 2	3	R6, R7, R8	3	7
C	2	R1, R2, R3	2	4	R7, R8	2, 2	6
D	6	R1, R2, R3, R4	3, 3	5	R8	5	11
E	3	R1, R2, R3, R4, R5	3	0	----------	0	3
F	8	R2, R3, R4, R5	3, 5	0	R6	0	8
G	13	R3, R4, R5	3, 5, 5	3	R6, R7	3	16
H	7	R4, R5	2, 5	7	R6, R7, R8	2, 5	14

Step 3:

From condition (i), team F scored 5 and 2 points in rounds R2 and R4 respectively.

From condition (ii), both teams D and F scored in round R2.

From condition (iii), all teams scored in a maximum of two rounds out of R6, R7 and R8 rounds.

From condition (iv), only team C and H scored in two consecutive rounds from rounds 6 to 8.

From the above table, team B scored 2, 2 points in rounds R1 and R2. So team A scored 5, 2 points in rounds R1 and R5 respectively. Therefore, C scored 2 points in round R3.

So, team H scored 2, 5 points in rounds R4 and R5 respectively. Hence, team G scored 3, 5, 5 points in rounds R5, R4 and R3 respectively. Similarly, we can fill for the other teams.

Hence, the final table can be shown as:

Round No.	5 Points	3 Points	2 Points
R1	A	D/E	B
R2	F	D	B
R3	G	E/D	C
R4	G	F	H
R5	H	G	A
R6	A	B/G	H
R7	H	G/B	C

R8	D	A	C

The scores of teams B, C, E, H were 4, 2, 3 and 0 respectively after Round 3.
So, teams B, C, E, H were in the last four positions.

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The following information is also known.
(i) Team F did not score consecutively in any two rounds from Rounds 1 to 5.
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(iii) No team scored in all the three rounds from Rounds 6 to 8.
(iv) Only two teams scored in two consecutive rounds from Rounds 6 to 8.

Q.3 [11831809]
Which of the following statements is/are definitely correct?
I. Team B scored in Rounds 1, 2 & 7.
II. Team E scored in Round 3.

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1 ◯ I only

---

2 ◯ II only

---

3 ◯ Both I & II

4 ◯ Neither I nor II

Solution:
Correct Answer : 4

Answer key/Solution

Let's say R1, R2, …, R8 represent the rounds 1 to 8.

Step 1:

From the given information, first we calculate the points scored by each team from rounds 6 to 8. Points scored by team A from rounds 6 to 8 = 15 – 7 = 8. Similarly, we can find for the other teams. Then, we find which teams participated in the given rounds. Team A participated in rounds R1, R5, R6, R7, R8. Similarly, we can find for other teams.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5		8	R6, R7, R8		15
B	4	R1, R2		3	R6, R7, R8		7
C	2	R1, R2, R3		4	R7, R8		6
D	6	R1, R2, R3, R4		5	R8		11
E	3	R1, R2, R3, R4, R5		0	----------		3
F	8	R2, R3, R4, R5		0	R6		8
G	13	R3, R4, R5		3	R6, R7		16
H	7	R4, R5		7	R6, R7, R8		14

Step 2:

Now let us calculate the points scored by the teams in different rounds.

From condition (i), team F scored in two rounds out of four rounds R2, R3, R4, R5.

From condition (iii), all teams scored in a maximum of two rounds from rounds 6 to 8.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
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D	6	R1, R2, R3, R4	3, 3	5	R8	5	11
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F	8	R2, R3, R4, R5	3, 5	0	R6	0	8
G	13	R3, R4, R5	3, 5, 5	3	R6, R7	3	16
H	7	R4, R5	2, 5	7	R6, R7, R8	2, 5	14

Step 3:

From condition (i), team F scored 5 and 2 points in rounds R2 and R4 respectively.

From condition (ii), both teams D and F scored in round R2.

From condition (iii), all teams scored in a maximum of two rounds out of R6, R7 and R8 rounds.

From condition (iv), only team C and H scored in two consecutive rounds from rounds 6 to 8.

From the above table, team B scored 2, 2 points in rounds R1 and R2. So team A scored 5, 2 points in rounds R1 and R5 respectively. Therefore, C scored 2 points in round R3.

So, team H scored 2, 5 points in rounds R4 and R5 respectively. Hence, team G scored 3, 5, 5 points in rounds R5, R4 and R3 respectively. Similarly, we can fill for the other teams.

Hence, the final table can be shown as:

Round No.	5 Points	3 Points	2 Points
R1	A	D/E	B
R2	F	D	B
R3	G	E/D	C
R4	G	F	H
R5	H	G	A
R6	A	B/G	H
R7	H	G/B	C

R8	D	A	C

Neither statement I nor statement II is definitely correct.

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Directions for questions 1 to 6: Answer the questions on the basis of the information given below.

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The top three performing teams in each round were awarded 5, 3 and 2 points respectively. There was no tie in any of the 8 rounds. The table below shows the total points scored by 8 teams after Round 5 and Round 8.

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H	7	14

The following information is also known.
(i) Team F did not score consecutively in any two rounds from Rounds 1 to 5.
(ii) Both teams D and F scored in a round from Rounds 1 to 5.
(iii) No team scored in all the three rounds from Rounds 6 to 8.
(iv) Only two teams scored in two consecutive rounds from Rounds 6 to 8.

Q.4 [11831809]
How many teams did not score 3 points in any of the 8 rounds?

Solution:
Correct Answer : 2

Answer key/Solution

Let's say R1, R2, …, R8 represent the rounds 1 to 8.

Step 1:

From the given information, first we calculate the points scored by each team from rounds 6 to 8. Points scored by team A from rounds 6 to 8 = 15 – 7 = 8. Similarly, we can find for the other teams. Then, we find which teams participated in the given rounds. Team A participated in rounds R1, R5, R6, R7, R8. Similarly, we can find for other teams.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5		8	R6, R7, R8		15
B	4	R1, R2		3	R6, R7, R8		7
C	2	R1, R2, R3		4	R7, R8		6
D	6	R1, R2, R3, R4		5	R8		11
E	3	R1, R2, R3, R4, R5		0	----------		3
F	8	R2, R3, R4, R5		0	R6		8
G	13	R3, R4, R5		3	R6, R7		16
H	7	R4, R5		7	R6, R7, R8		14

Step 2:

Now let us calculate the points scored by the teams in different rounds.

From condition (i), team F scored in two rounds out of four rounds R2, R3, R4, R5.

From condition (iii), all teams scored in a maximum of two rounds from rounds 6 to 8.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5	2, 5	8	R6, R7, R8	3, 5	15
B	4	R1, R2	2, 2	3	R6, R7, R8	3	7
C	2	R1, R2, R3	2	4	R7, R8	2, 2	6
D	6	R1, R2, R3, R4	3, 3	5	R8	5	11
E	3	R1, R2, R3, R4, R5	3	0	----------	0	3
F	8	R2, R3, R4, R5	3, 5	0	R6	0	8
G	13	R3, R4, R5	3, 5, 5	3	R6, R7	3	16
H	7	R4, R5	2, 5	7	R6, R7, R8	2, 5	14

Step 3:

From condition (i), team F scored 5 and 2 points in rounds R2 and R4 respectively.

From condition (ii), both teams D and F scored in round R2.

From condition (iii), all teams scored in a maximum of two rounds out of R6, R7 and R8 rounds.

From condition (iv), only team C and H scored in two consecutive rounds from rounds 6 to 8.

From the above table, team B scored 2, 2 points in rounds R1 and R2. So team A scored 5, 2 points in rounds R1 and R5 respectively. Therefore, C scored 2 points in round R3.

So, team H scored 2, 5 points in rounds R4 and R5 respectively. Hence, team G scored 3, 5, 5 points in rounds R5, R4 and R3 respectively. Similarly, we can fill for the other teams.

Hence, the final table can be shown as:

Round No.	5 Points	3 Points	2 Points
R1	A	D/E	B
R2	F	D	B
R3	G	E/D	C
R4	G	F	H
R5	H	G	A
R6	A	B/G	H
R7	H	G/B	C

R8	D	A	C

Two teams C and H did not score 3 points in any of the 8 rounds.

Bookmark FeedBack

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The following information is also known.
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(iii) No team scored in all the three rounds from Rounds 6 to 8.
(iv) Only two teams scored in two consecutive rounds from Rounds 6 to 8.

Q.5 [11831809]
If the team that scored in maximum number of rounds could win the Quiz, then which of the following teams definitely did not win the Quiz?

1 ○ A

2 ○ F

3 ○ G

4 ○ H

Solution:
Correct Answer : 2

Answer key/Solution

Let's say R1, R2, …, R8 represent the rounds 1 to 8.

Step 1:

From the given information, first we calculate the points scored by each team from rounds 6 to 8. Points scored by team A from rounds 6 to 8 = 15 – 7 = 8. Similarly, we can find for the other teams. Then, we find which teams participated in the given rounds. Team A participated in rounds R1, R5, R6, R7, R8. Similarly, we can find for other teams.

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D	6	R1, R2, R3, R4		5	R8		11
E	3	R1, R2, R3, R4, R5		0	----------		3
F	8	R2, R3, R4, R5		0	R6		8
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Step 2:

Now let us calculate the points scored by the teams in different rounds.

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D	6	R1, R2, R3, R4	3, 3	5	R8	5	11
E	3	R1, R2, R3, R4, R5	3	0	----------	0	3
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Step 3:

From condition (i), team F scored 5 and 2 points in rounds R2 and R4 respectively.

From condition (ii), both teams D and F scored in round R2.

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From the above table, team B scored 2, 2 points in rounds R1 and R2. So team A scored 5, 2 points in rounds R1 and R5 respectively. Therefore, C scored 2 points in round R3.

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Hence, the final table can be shown as:

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R2	F	D	B
R3	G	E/D	C
R4	G	F	H
R5	H	G	A
R6	A	B/G	H
R7	H	G/B	C

R8	D	A	C

Since team F scored in only three rounds. So, team F definitely did not win the Quiz.

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(iv) Only two teams scored in two consecutive rounds from Rounds 6 to 8.

Q.6 [11831809]
Which of the following teams did not score in Round 8?

1 ○ A

2 ○ C

3 ○ D

4 ○ H

Solution:
Correct Answer : 4

Answer key/Solution

Let's say R1, R2, …, R8 represent the rounds 1 to 8.

Step 1:

From the given information, first we calculate the points scored by each team from rounds 6 to 8. Points scored by team A from rounds 6 to 8 = 15 – 7 = 8. Similarly, we can find for the other teams. Then, we find which teams participated in the given rounds. Team A participated in rounds R1, R5, R6, R7, R8. Similarly, we can find for other teams.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5		8	R6, R7, R8		15
B	4	R1, R2		3	R6, R7, R8		7
C	2	R1, R2, R3		4	R7, R8		6
D	6	R1, R2, R3, R4		5	R8		11
E	3	R1, R2, R3, R4, R5		0	----------		3
F	8	R2, R3, R4, R5		0	R6		8
G	13	R3, R4, R5		3	R6, R7		16
H	7	R4, R5		7	R6, R7, R8		14

Step 2:

Now let us calculate the points scored by the teams in different rounds.

From condition (i), team F scored in two rounds out of four rounds R2, R3, R4, R5.

From condition (iii), all teams scored in a maximum of two rounds from rounds 6 to 8.

Team	Scores after round 5	Round No.	Points scored	Scores from round 6 to 8	Round No.	Points scored	Scores after round 8
A	7	R1, R5	2, 5	8	R6, R7, R8	3, 5	15
B	4	R1, R2	2, 2	3	R6, R7, R8	3	7
C	2	R1, R2, R3	2	4	R7, R8	2, 2	6
D	6	R1, R2, R3, R4	3, 3	5	R8	5	11
E	3	R1, R2, R3, R4, R5	3	0	----------	0	3
F	8	R2, R3, R4, R5	3, 5	0	R6	0	8
G	13	R3, R4, R5	3, 5, 5	3	R6, R7	3	16
H	7	R4, R5	2, 5	7	R6, R7, R8	2, 5	14

Step 3:

From condition (i), team F scored 5 and 2 points in rounds R2 and R4 respectively.

From condition (ii), both teams D and F scored in round R2.

From condition (iii), all teams scored in a maximum of two rounds out of R6, R7 and R8 rounds.

From condition (iv), only team C and H scored in two consecutive rounds from rounds 6 to 8.

From the above table, team B scored 2, 2 points in rounds R1 and R2. So team A scored 5, 2 points in rounds R1 and R5 respectively. Therefore, C scored 2 points in round R3.

So, team H scored 2, 5 points in rounds R4 and R5 respectively. Hence, team G scored 3, 5, 5 points in rounds R5, R4 and R3 respectively. Similarly, we can fill for the other teams.

Hence, the final table can be shown as:

Round No.	5 Points	3 Points	2 Points
R1	A	D/E	B
R2	F	D	B
R3	G	E/D	C
R4	G	F	H
R5	H	G	A
R6	A	B/G	H
R7	H	G/B	C

R8	D	A	C

Team H did not score in Round 8.

Bookmark FeedBack

Directions for questions 7 to 10: Answer the questions on the basis of the information given below.

An artist will paint 15 tiles on the floor in the colors - green, red, and/or purple. The tiles are arranged in three symmetrical columns as shown in the following diagram:

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

The artist tells you that she will paint each tile a certain color according to the following conditions:
(i) The corner tiles are painted red.
(ii) Tile 7 is painted red.
(iii) Every tile that borders atleast two red tiles must be a green tile. A tile borders another tile if they have an edge in common.
(iv) Every green tile must border a purple tile.

Q.7 [11831809]
If tile 9 is red in colour, then how many purple tiles are there?

Solution:
Correct Answer : 2

Answer key/Solution

Given:

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Since every tile that borders two red tiles must be a green tile, therefore,

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Also, every green tile must border a purple tile.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Let tile 9 be red in colour, then as per condition (iii), "every tile that borders atleast two red tiles must be a green tile", which implies, tile 6, 8 and 12 must be green and so, only 2 tiles 5 and 11 will be purple-coloured tiles.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Bookmark FeedBack

Directions for questions 7 to 10: Answer the questions on the basis of the information given below.

An artist will paint 15 tiles on the floor in the colors - green, red, and/or purple. The tiles are arranged in three symmetrical columns as shown in the following diagram:

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

The artist tells you that she will paint each tile a certain color according to the following conditions:
(i) The corner tiles are painted red.
(ii) Tile 7 is painted red.
(iii) Every tile that borders atleast two red tiles must be a green tile. A tile borders another tile if they have an edge in common.
(iv) Every green tile must border a purple tile.

Q.8 [11831809]
Which colour has the most possible tiles?

1 ◯ Green

2 ◯ Purple

3 ◯ Red

4 ◯ Either Green or Red

Solution:
Correct Answer : 4

Answer key/Solution

Given:

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Since every tile that borders two red tiles must be a green tile, therefore,

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Also, every green tile must border a purple tile.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

To find the colour which has the most possible tiles observe that as per incomplete structure above red, green and purple have 5, 4, and 2 tiles respectively. Also, there can be atmost two other red-coloured tiles, so purple is definitely not the answer.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Also, there can be atmost three other green-coloured tiles, i.e., total 7 tiles. Therefore, either Red or Green can have maximum possible number of tiles.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Bookmark FeedBack

Directions for questions 7 to 10: Answer the questions on the basis of the information given below.

An artist will paint 15 tiles on the floor in the colors - green, red, and/or purple. The tiles are arranged in three symmetrical columns as shown in the following diagram:

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

The artist tells you that she will paint each tile a certain color according to the following conditions:
(i) The corner tiles are painted red.
(ii) Tile 7 is painted red.
(iii) Every tile that borders atleast two red tiles must be a green tile. A tile borders another tile if they have an edge in common.
(iv) Every green tile must border a purple tile.

Q.9 [11831809]
How many different possible combinations of coloured tiles can be there, if there are exactly 6 red tiles?

1 ◯ 21

2 ◯ 22

3 ◯ 23

4 ◯ 28

Solution:
Correct Answer : 2

Answer key/Solution

Given:

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Since every tile that borders two red tiles must be a green tile, therefore,

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Also, every green tile must border a purple tile.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Given: There are exactly 6 red tiles.
By default, red already have 5 tiles at present. Only one more has to be arranged in four vacant spaces.
Case 1: Let Tile 6 be red.
Subcase 1: 8, 9, 11 and 12 tile can all be purple.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

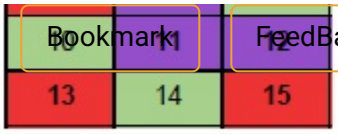
Subcase 2:
1. Tile 8 - green, Tile 9 and 12 - purple.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

2. Tile 9 - green, tiles 8 and 12 - purple.

1	2	3
4	5	6
7	8	9

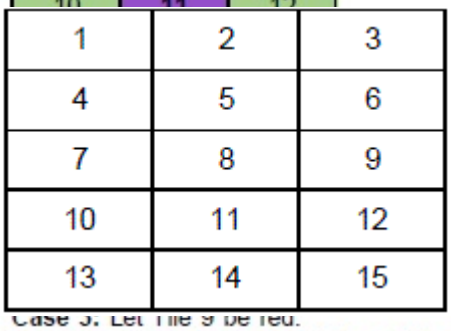
Bookmark FeedBack



3. Tile 12 - green, tiles 8 and 9 - purple.

Directions for questions 7 to 10: Answer the questions on the basis of the information given below.

An artist will paint 15 tiles on the floor in the colors - green, red, and/or purple. The tiles are arranged in three symmetrical columns as shown in the following diagram:



The artist tells you that she will paint each tile a certain color according to the following conditions:
(i) The corner tiles are painted red.
(ii) Tile 7 is painted red.
(iii) Every tile that borders atleast two red tiles must be a green tile. A tile borders another tile if they have an edge in common.
(iv) Every green tile must border a purple tile.

Q.10 [11831809]

If a purple tile borders exactly one green tile, then it must border a red tile too. If there are exactly four purple-coloured tiles, then which of the following statements is DEFINITELY false?

1 ◯ There is exactly one even colored red tile.

2 ◯ If tile 6 is purple, tile 9 is Green.

3 ◯ If tile 12 is purple, tile 8 is Red.

4 ◯ Sum of all purple-colored tiles is always either 36 or 33.

Answer key/Solution

Solution:

Correct Answer : 4

Given:

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Since every tile that borders two red tiles must be a green tile, therefore,

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Also, every green tile must border a purple tile.

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

Observe the options:

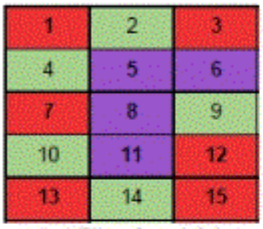
Option 1: Let us take the contradiction that two even colored red tiles are there: (6, 8), (6, 12) or (8, 12).

In all the cases, tile (9 and 5), tile 9 and tile (9 and 11) must be green,
but that will be false since those tiles will already be purple.

Hence, one out of 6, 8 or 12 can be red.

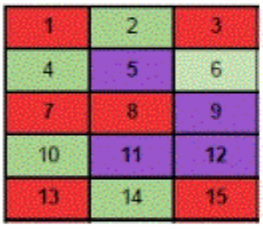
Option 2: If tile 6 is purple, then tile 9 is green is a possible case.

So therefore, it's not definitely false.



Option 3: If tile 12 is purple, then tile 8 is red is a possible case.

So therefore, it's not definitely false.



Option 4: Tiles 5 and 11 are already purple.

Also, as per the question statement exactly four tiles are purple.

Other possible combinations of remaining purple tiles are:

(6, 8), (6, 9), (6, 12), (8, 9), (8, 12), and (9, 12).

Possible cases	Sum
(6, 8)	30
(6, 9)	31
(6, 12)	34
(8, 9)	33
(8, 12)	36
(9, 12)	37

Therefore, sum is not always 36 or 33. Hence, this statement is definitely false.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

At an Indian wedding the guests were served two full meals - lunch and dinner - on the same day. Each meal had 3 courses - starter, main course and dessert. There were six caterers - A, B, C, D, E and F - who were appointed for the preparation of these meals. In each of the three courses the dinner menu had more dishes than the lunch menu. Also the main course did not have a lesser number of dishes than the starter or dessert menu. Considering lunch and dinner together each caterer prepared and served exactly four dishes. Each caterer served at least one dish during each meal. The guests were served one dish at a time in order. The starters preceded the main course, which preceded the desserts. The number of dishes in the starters menu and the main course during dinner was exactly one more than those in the lunch menu. Dinner had more desserts than starters. There were at least 3 starters and 3 desserts in each meal. Dinner had four more dishes than lunch. None of the caterers served more than one dish in any course. Each caterer prepared at least one starter and one dessert on the wedding day. The time taken for preparation of each dish in the starter, main course and dessert menus during lunch was 1 hour, 2 hours and 1.5 hours respectively whereas during dinner the preparation time was 30 minutes more for each dish in the respective courses. Any caterer does not prepare more than one dish at a time.

Further, it is also known that:
(i) B served the 1st and last dishes on the lunch menu as well as the dinner menu.
(ii) A served the 3rd and 4th dishes on the lunch menu and the 9th and 10th dishes on the dinner menu. C served two dishes during dinner, one immediatly before and the other immediatly after A.
(iii) F served the 7th and 8th dishes on the lunch menu and the 4th and 5th dishes in the dinner menu.
(iv) E's dish always follows D's dish in any course.

Q.11 [11831809]
The second dessert in the lunch menu was prepared by

1 ◯ A

2 ◯ F

3 ◯ C

4 ◯ E

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Let S_i, M_i and D_i represent the starter, main course and dessert dishes.
Six caterers served 4 dishes each so there were 24 dishes served in all. Also, the dinner menu had 4 more dishes than the lunch menu, so lunch had 10 and dinner had 14 dishes.
In each meal there were at least 3 starters and 3 desserts served and the main course had at least as many dishes as the dessert menu. So for lunch the starter, main course and dessert menu had 3, 4 and 3 dishes each. Also dinner had more desserts than starters, so the same courses in the dinner menu had 4, 5 and 5 courses each.
Step 2:
Using (i), (ii), (iii) and (iv) we can complete the following table:

S. No.	Lunch			Dinner		
	Course	Caterer	Prepared time (in hour)	Course	Caterer	Prepared time (in hour)
1	S1	B	1	S1	B	1.5
2	S2	C	1	S2	D	1.5
3	S3	A	1	S3	E	1.5
4	M1	A	2	S4	F	1.5
5	M2	D	2	M1	F	2.5
6	M3	E	2	M2	D	2.5
7	M4	F	2	M3	E	2.5
8	D1	F	1.5	M4	C	2.5
9	D2	C	1.5	M5	A	2.5
10	D3	B	1.5	D1	A	2
11				D2	C	2
12				D3	D	2
13				D4	E	2
14				D5	B	2

Step 3:
Total time (in hours) taken by each caterer can be shown as:

A	B	C	D	E	F
1	1	1	2	2	2
2	1.5	1.5	1.5	1.5	1.5
2.5	1.5	2.5	2.5	2.5	1.5
2	2	2	2	2	2.5
7.5	6	7	8	8	7.5

The second dessert in the lunch menu was prepared by C.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

At an Indian wedding the guests were served two full meals - lunch and dinner - on the same day. Each meal had 3 courses - starter, main course and dessert. There were six caterers - A, B, C, D, E and F - who were appointed for the preparation of these meals. In each of the three courses the dinner menu had more dishes than the lunch menu. Also the main course did not have a lesser number of dishes than the starter or dessert menu. Considering lunch and dinner together each caterer prepared and served exactly four dishes. Each caterer served at least one dish during each meal. The guests were served one dish at a time in order. The starters preceded the main course, which preceded the desserts. The number of dishes in the starters menu and the main course during dinner was exactly one more than those in the lunch menu. Dinner had more desserts than starters. There were at least 3 starters and 3 desserts in each meal. Dinner had four more dishes than lunch. None of the caterers served more than one dish in any course. Each caterer prepared at least one starter and one dessert on the wedding day. The time taken for preparation of each dish in the starter, main course and dessert menus during lunch was 1 hour, 2 hours and 1.5 hours respectively whereas during dinner the preparation time was 30 minutes more for each dish in the respective courses. Any caterer does not prepare more than one dish at a time.

Further, it is also known that:
(i) B served the 1st and last dishes on the lunch menu as well as the dinner menu.
(ii) A served the 3rd and 4th dishes on the lunch menu and the 9th and 10th dishes on the dinner menu. C served two dishes during dinner, one immediatly before and the other immediatly after A.
(iii) F served the 7th and 8th dishes on the lunch menu and the 4th and 5th dishes in the dinner menu.
(iv) E's dish always follows D's dish in any course.

Q.12 [11831809]
Which of the following statements is incorrect based on the given information?

1 ◯ A and F spend the same amount of time in food preparation.

2 ◯ B spends more time than C in food preparation.

3 ◯ D and E spend 8 hours each in food preparation.

4 ◯ E spends more time than A in food preparation.

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
Let S_i, M_i and D_i represent the starter, main course and dessert dishes.
Six caterers served 4 dishes each so there were 24 dishes served in all. Also, the dinner menu had 4 more dishes than the lunch menu, so lunch had 10 and dinner had 14 dishes.
In each meal there were at least 3 starters and 3 desserts served and the main course had at least as many dishes as the dessert menu. So for lunch the starter, main course and dessert menu had 3, 4 and 3 dishes each. Also dinner had more desserts than starters, so the same courses in the dinner menu had 4, 5 and 5 courses each.

Step 2:
Using (i), (ii), (iii) and (iv) we can complete the following table:

S. No.	Lunch			Dinner		
	Course	Caterer	Prepared time (in hour)	Course	Caterer	Prepared time (in hour)
1	S1	B	1	S1	B	1.5
2	S2	C	1	S2	D	1.5
3	S3	A	1	S3	E	1.5
4	M1	A	2	S4	F	1.5
5	M2	D	2	M1	F	2.5
6	M3	E	2	M2	D	2.5
7	M4	F	2	M3	E	2.5
8	D1	F	1.5	M4	C	2.5
9	D2	C	1.5	M5	A	2.5
10	D3	B	1.5	D1	A	2
11				D2	C	2
12				D3	D	2
13				D4	E	2
14				D5	B	2

Step 3:
Total time (in hours) taken by each caterer can be shown as:

A	B	C	D	E	F
1	1	1	2	2	2
2	1.5	1.5	1.5	1.5	1.5
2.5	1.5	2.5	2.5	2.5	1.5
2	2	2	2	2	2.5
7.5	6	7	8	8	7.5

It is clear from the given information that option (2) is incorrect, as B spent 6 hours whereas C spent 7 hours in food preparation.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

At an Indian wedding the guests were served two full meals - lunch and dinner - on the same day. Each meal had 3 courses - starter, main course and dessert. There were six caterers - A, B, C, D, E and F - who were appointed for the preparation of these meals. In each of the three courses the dinner menu had more dishes than the lunch menu. Also the main course did not have a lesser number of dishes than the starter or dessert menu. Considering lunch and dinner together each caterer prepared and served exactly four dishes. Each caterer served at least one dish during each meal. The guests were served one dish at a time in order. The starters preceded the main course, which preceded the desserts. The number of dishes in the starters menu and the main course during dinner was exactly one more than those in the lunch menu. Dinner had more desserts than starters. There were at least 3 starters and 3 desserts in each meal. Dinner had four more dishes than lunch. None of the caterers served more than one dish in any course. Each caterer prepared at least one starter and one dessert on the wedding day. The time taken for preparation of each dish in the starter, main course and dessert menus during lunch was 1 hour, 2 hours and 1.5 hours respectively whereas during dinner the preparation time was 30 minutes more for each dish in the respective courses. Any caterer does not prepare more than one dish at a time.

Further, it is also known that:
(i) B served the 1st and last dishes on the lunch menu as well as the dinner menu.
(ii) A served the 3rd and 4th dishes on the lunch menu and the 9th and 10th dishes on the dinner menu. C served two dishes during dinner, one immediatly before and the other immediatly after A.
(iii) F served the 7th and 8th dishes on the lunch menu and the 4th and 5th dishes in the dinner menu.
(iv) E's dish always follows D's dish in any course.

Q.13 [11831809]
Who did not prepare dessert for the dinner menu?

1 ◯ F

2 ◯ A

3 ◯ B

4 ◯ D

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
Let S_i, M_i and D_i represent the starter, main course and dessert dishes.
Six caterers served 4 dishes each so there were 24 dishes served in all. Also, the dinner menu had 4 more dishes than the lunch menu, so lunch had 10 and dinner had 14 dishes.
In each meal there were at least 3 starters and 3 desserts served and the main course had at least as many dishes as the dessert menu. So for lunch the starter, main course and dessert menu had 3, 4 and 3 dishes each. Also dinner had more desserts than starters, so the same courses in the dinner menu had 4, 5 and 5 courses each.

Step 2:
Using (i), (ii), (iii) and (iv) we can complete the following table:

S. No.	Lunch			Dinner		
	Course	Caterer	Prepared time (in hour)	Course	Caterer	Prepared time (in hour)
1	S1	B	1	S1	B	1.5
2	S2	C	1	S2	D	1.5
3	S3	A	1	S3	E	1.5
4	M1	A	2	S4	F	1.5
5	M2	D	2	M1	F	2.5
6	M3	E	2	M2	D	2.5
7	M4	F	2	M3	E	2.5
8	D1	F	1.5	M4	C	2.5
9	D2	C	1.5	M5	A	2.5
10	D3	B	1.5	D1	A	2
11				D2	C	2
12				D3	D	2
13				D4	E	2
14				D5	B	2

Step 3:
Total time (in hours) taken by each caterer can be shown as:

A	B	C	D	E	F
1	1	1	2	2	2
2	1.5	1.5	1.5	1.5	1.5
2.5	1.5	2.5	2.5	2.5	1.5
2	2	2	2	2	2.5
7.5	6	7	8	8	7.5

It is clear from the table that F did not prepare dessert for the dinner menu.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

At an Indian wedding the guests were served two full meals - lunch and dinner - on the same day. Each meal had 3 courses - starter, main course and dessert. There were six caterers - A, B, C, D, E and F - who were appointed for the preparation of these meals. In each of the three courses the dinner menu had more dishes than the lunch menu. Also the main course did not have a lesser number of dishes than the starter or dessert menu. Considering lunch and dinner together each caterer prepared and served exactly four dishes. Each caterer served at least one dish during each meal. The guests were served one dish at a time in order. The starters preceded the main course, which preceded the desserts. The number of dishes in the starters menu and the main course during dinner was exactly one more than those in the lunch menu. Dinner had more desserts than starters. There were at least 3 starters and 3 desserts in each meal. Dinner had four more dishes than lunch. None of the caterers served more than one dish in any course. Each caterer prepared at least one starter and one dessert on the wedding day. The time taken for preparation of each dish in the starter, main course and dessert menus during lunch was 1 hour, 2 hours and 1.5 hours respectively whereas during dinner the preparation time was 30 minutes more for each dish in the respective courses. Any caterer does not prepare more than one dish at a time.

Further, it is also known that:
(i) B served the 1st and last dishes on the lunch menu as well as the dinner menu.
(ii) A served the 3rd and 4th dishes on the lunch menu and the 9th and 10th dishes on the dinner menu. C served two dishes during dinner, one immediatly before and the other immediatly after A.
(iii) F served the 7th and 8th dishes on the lunch menu and the 4th and 5th dishes in the dinner menu.
(iv) E's dish always follows D's dish in any course.

Q.14 [11831809]
How many times during both the meals did caterers E and F serve dishes consecutively, one after the other?

1 ◯ One

2 ◯ Three

3 ◯ Zero

4 ◯ Two

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
Let S_i, M_i and D_i represent the starter, main course and dessert dishes.
Six caterers served 4 dishes each so there were 24 dishes served in all. Also, the dinner menu had 4 more dishes than the lunch menu, so lunch had 10 and dinner had 14 dishes.
In each meal there were at least 3 starters and 3 desserts served and the main course had at least as many dishes as the dessert menu. So for lunch the starter, main course and dessert menu had 3, 4 and 3 dishes each. Also dinner had more desserts than starters, so the same courses in the dinner menu had 4, 5 and 5 courses each.

Step 2:
Using (i), (ii), (iii) and (iv) we can complete the following table:

S. No.	Lunch			Dinner		
	Course	Caterer	Prepared time (in hour)	Course	Caterer	Prepared time (in hour)
1	S1	B	1	S1	B	1.5
2	S2	C	1	S2	D	1.5
3	S3	A	1	S3	E	1.5
4	M1	A	2	S4	F	1.5
5	M2	D	2	M1	F	2.5
6	M3	E	2	M2	D	2.5
7	M4	F	2	M3	E	2.5
8	D1	F	1.5	M4	C	2.5
9	D2	C	1.5	M5	A	2.5
10	D3	B	1.5	D1	A	2
11				D2	C	2
12				D3	D	2
13				D4	E	2
14				D5	B	2

Step 3:
Total time (in hours) taken by each caterer can be shown as:

A	B	C	D	E	F
1	1	1	2	2	2
2	1.5	1.5	1.5	1.5	1.5
2.5	1.5	2.5	2.5	2.5	1.5
2	2	2	2	2	2.5
7.5	6	7	8	8	7.5

Caterers E and F serve dishes consecutively, one after the other once during the main course at lunch time and once during the starters at dinner time. Hence, the correct answer is two.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

At an Indian wedding the guests were served two full meals - lunch and dinner - on the same day. Each meal had 3 courses - starter, main course and dessert. There were six caterers - A, B, C, D, E and F - who were appointed for the preparation of these meals. In each of the three courses the dinner menu had more dishes than the lunch menu. Also the main course did not have a lesser number of dishes than the starter or dessert menu. Considering lunch and dinner together each caterer prepared and served exactly four dishes. Each caterer served at least one dish during each meal. The guests were served one dish at a time in order. The starters preceded the main course, which preceded the desserts. The number of dishes in the starters menu and the main course during dinner was exactly one more than those in the lunch menu. Dinner had more desserts than starters. There were at least 3 starters and 3 desserts in each meal. Dinner had four more dishes than lunch. None of the caterers served more than one dish in any course. Each caterer prepared at least one starter and one dessert on the wedding day. The time taken for preparation of each dish in the starter, main course and dessert menus during lunch was 1 hour, 2 hours and 1.5 hours respectively whereas during dinner the preparation time was 30 minutes more for each dish in the respective courses. Any caterer does not prepare more than one dish at a time.

Further, it is also known that:
(i) B served the 1st and last dishes on the lunch menu as well as the dinner menu.
(ii) A served the 3rd and 4th dishes on the lunch menu and the 9th and 10th dishes on the dinner menu. C served two dishes during dinner, one immediatly before and the other immediatly after A.
(iii) F served the 7th and 8th dishes on the lunch menu and the 4th and 5th dishes in the dinner menu.
(iv) E's dish always follows D's dish in any course.

Q.15 [11831809]
Which of the following dishes was prepared by D?

1 ◯ 9th dish for lunch

2 ◯ 6th dish for dinner

3 ◯ 3rd dish for dinner

4 ◯ 13th dish for dinner

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
Let S_i, M_i and D_i represent the starter, main course and dessert dishes.
Six caterers served 4 dishes each so there were 24 dishes served in all. Also, the dinner menu had 4 more dishes than the lunch menu, so lunch had 10 and dinner had 14 dishes.
In each meal there were at least 3 starters and 3 desserts served and the main course had at least as many dishes as the dessert menu. So for lunch the starter, main course and dessert menu had 3, 4 and 3 dishes each. Also dinner had more desserts than starters, so the same courses in the dinner menu had 4, 5 and 5 courses each.

Step 2:
Using (i), (ii), (iii) and (iv) we can complete the following table:

S. No.	Lunch			Dinner		
	Course	Caterer	Prepared time (in hour)	Course	Caterer	Prepared time (in hour)
1	S1	B	1	S1	B	1.5
2	S2	C	1	S2	D	1.5
3	S3	A	1	S3	E	1.5
4	M1	A	2	S4	F	1.5
5	M2	D	2	M1	F	2.5
6	M3	E	2	M2	D	2.5
7	M4	F	2	M3	E	2.5
8	D1	F	1.5	M4	C	2.5
9	D2	C	1.5	M5	A	2.5
10	D3	B	1.5	D1	A	2
11				D2	C	2
12				D3	D	2
13				D4	E	2
14				D5	B	2

Step 3:
Total time (in hours) taken by each caterer can be shown as:

A	B	C	D	E	F
1	1	1	2	2	2
2	1.5	1.5	1.5	1.5	1.5
2.5	1.5	2.5	2.5	2.5	1.5
2	2	2	2	2	2.5
7.5	6	7	8	8	7.5

It is clear from the table that the 6th dish on the dinner menu was prepared by D.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

At an Indian wedding the guests were served two full meals - lunch and dinner - on the same day. Each meal had 3 courses - starter, main course and dessert. There were six caterers - A, B, C, D, E and F - who were appointed for the preparation of these meals. In each of the three courses the dinner menu had more dishes than the lunch menu. Also the main course did not have a lesser number of dishes than the starter or dessert menu. Considering lunch and dinner together each caterer prepared and served exactly four dishes. Each caterer served at least one dish during each meal. The guests were served one dish at a time in order. The starters preceded the main course, which preceded the desserts. The number of dishes in the starters menu and the main course during dinner was exactly one more than those in the lunch menu. Dinner had more desserts than starters. There were at least 3 starters and 3 desserts in each meal. Dinner had four more dishes than lunch. None of the caterers served more than one dish in any course. Each caterer prepared at least one starter and one dessert on the wedding day. The time taken for preparation of each dish in the starter, main course and dessert menus during lunch was 1 hour, 2 hours and 1.5 hours respectively whereas during dinner the preparation time was 30 minutes more for each dish in the respective courses. Any caterer does not prepare more than one dish at a time.

Further, it is also known that:
(i) B served the 1st and last dishes on the lunch menu as well as the dinner menu.
(ii) A served the 3rd and 4th dishes on the lunch menu and the 9th and 10th dishes on the dinner menu. C served two dishes during dinner, one immediatly before and the other immediatly after A.
(iii) F served the 7th and 8th dishes on the lunch menu and the 4th and 5th dishes in the dinner menu.
(iv) E's dish always follows D's dish in any course.

Q.16 [11831809]

The total time spent by C for preparing dishes for dinner is what percent more than the total time spent by C for preparing dishes for lunch?

Solution:
Correct Answer : 80

Answer key/Solution

Step 1:
Let S_i, M_i and D_i represent the starter, main course and dessert dishes.
Six caterers served 4 dishes each so there were 24 dishes served in all. Also, the dinner menu had 4 more dishes than the lunch menu, so lunch had 10 and dinner had 14 dishes.
In each meal there were at least 3 starters and 3 desserts served and the main course had at least as many dishes as the dessert menu. So for lunch the starter, main course and dessert menu had 3, 4 and 3 dishes each. Also dinner had more desserts than starters, so the same courses in the dinner menu had 4, 5 and 5 courses each.

Step 2:
Using (i), (ii), (iii) and (iv) we can complete the following table:

S. No.	Lunch			Dinner		
	Course	Caterer	Prepared time (in hour)	Course	Caterer	Prepared time (in hour)
1	S1	B	1	S1	B	1.5
2	S2	C	1	S2	D	1.5
3	S3	A	1	S3	E	1.5
4	M1	A	2	S4	F	1.5
5	M2	D	2	M1	F	2.5
6	M3	E	2	M2	D	2.5
7	M4	F	2	M3	E	2.5
8	D1	F	1.5	M4	C	2.5
9	D2	C	1.5	M5	A	2.5
10	D3	B	1.5	D1	A	2
11				D2	C	2
12				D3	D	2
13				D4	E	2
14				D5	B	2

Step 3:
Total time (in hours) taken by each caterer can be shown as:

A	B	C	D	E	F
1	1	1	2	2	2
2	1.5	1.5	1.5	1.5	1.5
2.5	1.5	2.5	2.5	2.5	1.5
2	2	2	2	2	2.5
7.5	6	7	8	8	7.5

Total time spent by C for preparing dishes for dinner = 2.5 + 2 = 4.5 hours
Total time spent by C for preparing dishes for lunch = 1 + 1.5 = 2.5 hours
Hence, required percentage = (4.5 – 2.5)/2.5 × 100 = 80%.

Bookmark FeedBack

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

The five Hand Sanitizer manufacturing companies - Godrej, Himalaya, Khadi, Pee Safe and Resil - had a total market share of 50% in a country in 2020. The details regarding their market share distribution, selling price per liter and profitability (defined as the profit as a percentage of the revenue) for the year 2020 are given in the table below.

Company	Market Share distribution (%)	Selling Price (Rs.)/Liter	Profitability (%)
Godrej	30	320	20
Himalaya	15	400	15
Khadi	20	460	25
Pee Safe	15	260	10
Resil	20	420	15

In 2021, the sales volume of hand sanitizers increased by 20% compared to 2020. The total market share (%) of these 5 companies remained the same in 2021 as in 2020. All companies offered 25% discount on its selling price in 2021, resulting in an increase of 5 percentage points each in the market share of Himalaya, Khadi and Pee Safe. Godrej and Resil lost 10 and 5 percentage points market share respectively. However, in 2021 the profitability (%) of Godrej, Himalaya and Khadi declined by 1/4, 1/3 and 1/5 respectively of the 2020 values. The profitability (%) of the other two companies remained the same in 2021 as it was in 2020.

Q.17 [11831809]

Which of the following companies had the third highest revenue in 2020?

1 ◯ Godrej

2 ◯ Himalaya

3 ◯ Khadi

4 ◯ Resil

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
Let the total market share of Hand Sanitizers in 2020 in the country be 100x.
Then, the total market share of given 5 companies = 100x × 0.5 = 50x
In 2020 for Godrej:
Share in the country = 15%
Share of Hand Sanitizers in the country = 15x
Revenue = 15x × 320 = Rs.4800x
Profit = 4800x × 0.2 = Rs.960x
Similarly, we can calculate for the other companies.
The details for 2020 can be shown in the following table:

Company	Share distribution within five (%)	Share distribution in the country	Selling Price (Rs.)/Liter	Revenue (Rs.)	Profitability (%)	Profit (Rs.)
Godrej	30	15x	320	4800x	20	960x
Himalaya	15	7.5x	400	3000x	15	450x
Khadi	20	10x	460	4600x	25	1150x
Pee Safe	15	7.5x	260	1950x	10	195x
Resil	20	10x	420	4200x	15	630x
	50%	50x				

Step 2:
The total market share of Hand Sanitizers in 2021 in the country = 100x × 1.2= 120x.
Then, the total market share of given 5 companies = 120x × 0.5 = 60x
The details for 2021 can be shown in the following table:

Company	Share distribution in the country (%)	Share distribution in the country	Selling Price (Rs.)/Liter	Revenue (Rs.)	Profitability (%)	Profit (Rs.)
Godrej	5	6x	240	1440x	15	216x
Himalaya	12.5	15x	300	4500x	10	450x
Khadi	15	18x	345	6210x	20	1242x
Pee Safe	12.5	15x	195	2925x	10	292.5x
Resil	5	6x	315	1890x	15	283.5x
	50%	60x				

Resil had the third highest revenue in 2020.

Bookmark FeedBack

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

The five Hand Sanitizer manufacturing companies - Godrej, Himalaya, Khadi, Pee Safe and Resil - had a total market share of 50% in a country in 2020. The details regarding their market share distribution, selling price per liter and profitability (defined as the profit as a percentage of the revenue) for the year 2020 are given in the table below.

Company	Market Share distribution (%)	Selling Price (Rs.)/Liter	Profitability (%)
Godrej	30	320	20
Himalaya	15	400	15
Khadi	20	460	25
Pee Safe	15	260	10
Resil	20	420	15

In 2021, the sales volume of hand sanitizers increased by 20% compared to 2020. The total market share (%) of these 5 companies remained the same in 2021 as in 2020. All companies offered 25% discount on its selling price in 2021, resulting in an increase of 5 percentage points each in the market share of Himalaya, Khadi and Pee Safe. Godrej and Resil lost 10 and 5 percentage points market share respectively. However, in 2021 the profitability (%) of Godrej, Himalaya and Khadi declined by 1/4, 1/3 and 1/5 respectively of the 2020 values. The profitability (%) of the other two companies remained the same in 2021 as it was in 2020.

Q.18 [11831809]

Which of the following companies had the second lowest profit in 2020?

1 ◯ Godrej

2 ◯ Pee Safe

3 ◯ Himalaya

4 ◯ Resil

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Let the total market share of Hand Sanitizers in 2020 in the country be 100x.
Then, the total market share of given 5 companies = 100x × 0.5 = 50x
In 2020 for Godrej:
Share in the country = 15%
Share of Hand Sanitizers in the country = 15x
Revenue = 15x × 320 = Rs.4800x
Profit = 4800x × 0.2 = Rs.960x
Similarly, we can calculate for the other companies.
The details for 2020 can be shown in the following table:

Company	Share distribution within five (%)	Share distribution in the country	Selling Price (Rs.)/Liter	Revenue (Rs.)	Profitability (%)	Profit (Rs.)
Godrej	30	15x	320	4800x	20	960x
Himalaya	15	7.5x	400	3000x	15	450x
Khadi	20	10x	460	4600x	25	1150x
Pee Safe	15	7.5x	260	1950x	10	195x
Resil	20	10x	420	4200x	15	630x
	50%	50x				

Step 2:
The total market share of Hand Sanitizers in 2021 in the country = 100x × 1.2= 120x.
Then, the total market share of given 5 companies = 120x × 0.5 = 60x
The details for 2021 can be shown in the following table:

Company	Share distribution in the country (%)	Share distribution in the country	Selling Price (Rs.)/Liter	Revenue (Rs.)	Profitability (%)	Profit (Rs.)
Godrej	5	6x	240	1440x	15	216x
Himalaya	12.5	15x	300	4500x	10	450x
Khadi	15	18x	345	6210x	20	1242x
Pee Safe	12.5	15x	195	2925x	10	292.5x
Resil	5	6x	315	1890x	15	283.5x
	50%	60x				

Himalaya had the second lowest profit in 2020.

Bookmark FeedBack

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

The five Hand Sanitizer manufacturing companies - Godrej, Himalaya, Khadi, Pee Safe and Resil - had a total market share of 50% in a country in 2020. The details regarding their market share distribution, selling price per liter and profitability (defined as the profit as a percentage of the revenue) for the year 2020 are given in the table below.

Company	Market Share distribution (%)	Selling Price (Rs.)/Liter	Profitability (%)
Godrej	30	320	20
Himalaya	15	400	15
Khadi	20	460	25
Pee Safe	15	260	10
Resil	20	420	15

In 2021, the sales volume of hand sanitizers increased by 20% compared to 2020. The total market share (%) of these 5 companies remained the same in 2021 as in 2020. All companies offered 25% discount on its selling price in 2021, resulting in an increase of 5 percentage points each in the market share of Himalaya, Khadi and Pee Safe. Godrej and Resil lost 10 and 5 percentage points market share respectively. However, in 2021 the profitability (%) of Godrej, Himalaya and Khadi declined by 1/4, 1/3 and 1/5 respectively of the 2020 values. The profitability (%) of the other two companies remained the same in 2021 as it was in 2020.

Q.19 [11831809]

The complete list of companies whose revenue grew from 2020 to 2021 was _____.

1 ◯ Himalaya, Khadi, Pee Safe

2 ◯ Godrej, Himalaya, Pee Safe

3 ◯ Khadi, Pee Safe, Resil

4 ◯ Himalaya, Khadi

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
Let the total market share of Hand Sanitizers in 2020 in the country be 100x.
Then, the total market share of given 5 companies = 100x × 0.5 = 50x
In 2020 for Godrej:
Share in the country = 15%
Share of Hand Sanitizers in the country = 15x
Revenue = 15x × 320 = Rs.4800x
Profit = 4800x × 0.2 = Rs.960x
Similarly, we can calculate for the other companies.
The details for 2020 can be shown in the following table:

Company	Share distribution within five (%)	Share distribution in the country	Selling Price (Rs.)/Liter	Revenue (Rs.)	Profitability (%)	Profit (Rs.)
Godrej	30	15x	320	4800x	20	960x
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Khadi	20	10x	460	4600x	25	1150x
Pee Safe	15	7.5x	260	1950x	10	195x
Resil	20	10x	420	4200x	15	630x
	50%	50x				

Step 2:
The total market share of Hand Sanitizers in 2021 in the country = 100x × 1.2= 120x.
Then, the total market share of given 5 companies = 120x × 0.5 = 60x
The details for 2021 can be shown in the following table:

Company	Share distribution in the country (%)	Share distribution in the country	Selling Price (Rs.)/Liter	Revenue (Rs.)	Profitability (%)	Profit (Rs.)
Godrej	5	6x	240	1440x	15	216x
Himalaya	12.5	15x	300	4500x	10	450x
Khadi	15	18x	345	6210x	20	1242x
Pee Safe	12.5	15x	195	2925x	10	292.5x
Resil	5	6x	315	1890x	15	283.5x
	50%	60x				

The companies whose revenue grew from 2020 to 2021 were Himalaya, Khadi and Pee Safe.

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Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

The five Hand Sanitizer manufacturing companies - Godrej, Himalaya, Khadi, Pee Safe and Resil - had a total market share of 50% in a country in 2020. The details regarding their market share distribution, selling price per liter and profitability (defined as the profit as a percentage of the revenue) for the year 2020 are given in the table below.

Company	Market Share distribution (%)	Selling Price (Rs.)/Liter	Profitability (%)
Godrej	30	320	20
Himalaya	15	400	15
Khadi	20	460	25
Pee Safe	15	260	10
Resil	20	420	15

In 2021, the sales volume of hand sanitizers increased by 20% compared to 2020. The total market share (%) of these 5 companies remained the same in 2021 as in 2020. All companies offered 25% discount on its selling price in 2021, resulting in an increase of 5 percentage points each in the market share of Himalaya, Khadi and Pee Safe. Godrej and Resil lost 10 and 5 percentage points market share respectively. However, in 2021 the profitability (%) of Godrej, Himalaya and Khadi declined by 1/4, 1/3 and 1/5 respectively of the 2020 values. The profitability (%) of the other two companies remained the same in 2021 as it was in 2020.

Q.20 [11831809]

The profit of how many companies did not increase from 2020 to 2021?

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Let the total market share of Hand Sanitizers in 2020 in the country be 100x.
Then, the total market share of given 5 companies = 100x × 0.5 = 50x
In 2020 for Godrej:
Share in the country = 15%
Share of Hand Sanitizers in the country = 15x
Revenue = 15x × 320 = Rs.4800x
Profit = 4800x × 0.2 = Rs.960x
Similarly, we can calculate for the other companies.
The details for 2020 can be shown in the following table:

Company	Share distribution within five (%)	Share distribution in the country	Selling Price (Rs.)/Liter	Revenue (Rs.)	Profitability (%)	Profit (Rs.)
Godrej	30	15x	320	4800x	20	960x
Himalaya	15	7.5x	400	3000x	15	450x
Khadi	20	10x	460	4600x	25	1150x
Pee Safe	15	7.5x	260	1950x	10	195x
Resil	20	10x	420	4200x	15	630x
	50%	50x				

Step 2:
The total market share of Hand Sanitizers in 2021 in the country = 100x × 1.2= 120x.
Then, the total market share of given 5 companies = 120x × 0.5 = 60x
The details for 2021 can be shown in the following table:

Company	Share distribution in the country (%)	Share distribution in the country	Selling Price (Rs.)/Liter	Revenue (Rs.)	Profitability (%)	Profit (Rs.)
Godrej	5	6x	240	1440x	15	216x
Himalaya	12.5	15x	300	4500x	10	450x
Khadi	15	18x	345	6210x	20	1242x
Pee Safe	12.5	15x	195	2925x	10	292.5x
Resil	5	6x	315	1890x	15	283.5x
	50%	60x				

The profit of 3 companies (Godrej, Himalaya & Resil) did not increase from 2020 to 2021.

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